NORMALISATION (TASK 2)

UNF

COLLECTION_REPORT (contract_no, waste_type_id, waste_type_desc,
collection_frequency, collect_date, truck_vin, truck_rego,
truck_make, truck_model, truck_year,
(bin_rfid, collected_kgs, overweight))

1NF

COLLECTION_REPORT (contract_no, waste_type_id, waste_type_desc,
collection_frequency, collect_date, truck_vin, truck_rego,
truck_make, truck_model, truck_year, bin_rfid,
collected_kgs, overweight)

Candidate Keys:

- CK: contract_no, collect_date, bin_rfid
- CK: truck_vin, collect_date, bin_rfid

Partial dependencies:

- contract_no → waste_type_id, waste_type_desc, collection_frequency
- collect_date , contract_no → truck_vin, truck_rego, truck_make, truck_model,truck_year

2NF

CONTRACT_WASTE (contract_no, waste_type_id, waste_type_desc,
collection_frequency)
CK: contract_no

COLLECTION (contract_no, collect_date, truck_vin, truck_rego,
truck_make, truck_model, truck_year)

CK: contract_no, collect_date
CK: truck_vin, collect_date

BIN_COLLECTION (contract_no, collect_date, bin_rfid,
collected_kgs, overweight)

CK: contract_no, collect_date, bin_rfid

Transitive dependencies:
waste_type_id → waste_type_desc
truck_vin → truck_rego, truck_make, truck_model, truck_year

3NF

CONTRACT_WASTE (contract_no, waste_type_id, collection_frequency)
CK: contract_no

WASTE_TYPE (waste_type_id, waste_type_desc)
CK: waste_type_id

COLLECTION (contract_no, collect_date, truck_vin)
CK: contract_no, collect_date
CK: truck_vin, collect_date

TRUCK (truck_vin, truck_rego, truck_make, truck_model, truck_year)
CK: truck_vin
CK: truck_rego

BIN_COLLECTION (contract_no, collect_date, bin_rfid, collected_kgs, overweight)
CK: contract_no, collect_date, bin_rfid

Full dependencies
contract_no → waste_type_id, collection_frequency
waste_type_id → waste_type_desc
contract_no, collect_date → truck_vin
truck_vin → truck_rego, truck_make, truck_model, truck_year
contract_no, collect_date, bin_rfid → collected_kgs, overweight

Attribute Synthesis:

No synthesis required , no duplicate relation structures identified.